Position Self-Sensing for Electrostatic Motors Under Amplitude-Modulated AC Operation

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***Abstract*—In a previous study, displacement self-sensing using driving current has been proposed for a synchronous-type macroscale electrostatic motor under alternating current (ac) operation; however, it cannot be used in practice due to its low noise immunity. This work improves the position self-sensing method for electrostatic motors under amplitude-modulated ac operation, in which the driving signal is amplitude-modulated using a high-frequency carrier signal. The modulation transfers the signal frequency to a higher region, which results in a much larger driving current and high noise immunity. In the modified self-sensing method discussed in this work, the high-frequency signal is demodulated, and then the slider displacement information is distilled through Clarke and Park transformations. The systematic error of the method is discussed to show that it depends on the performance of the cutoff frequency of the low-pass filter (LPF) in demodulation. Theoretical analysis shows that the accuracy is possibly improved by decreasing the cutoff frequency of the LPF. Simulation results confirmed that the noise immunity of this modified self-sensing method is much higher than that of the previous method, which means it can be used in practice. Verification experiments have been done under various conditions by using prototype electrostatic motors, which validated that the modified self-sensing method realizes a high accuracy/resolution of around 20 μm in practical situations.**

***Index Terms*—Electrostatic motors, position self-sensing, synchronous machines.**

I. Introduction

ELECTROSTATIC motors have been developed for over two hundred years [1], and various different motors were conceived, mainly including corona motor [2], [3], electrostatic micromotor [4], [5], macroscale electrostatic motor [6]. In a corona motor, high voltage is applied to the stator electrodes/blades with sharp edges to ionize air molecules and spray the ionized air to the rotor surface, such that the rotor is attracted/repelled by the stator blades [7]. Electrostatic micromotor has been widely adopted for providing actuation force in microelectromechanical systems (MEMS) and microrobot systems due to its high force density in microscale structure [8], [9], [10], [11]. Macroscale electrostatic motors share the basic driving principles with electrostatic micromotors. They had not been competitive with electromagnetic motors in the past. However, since printed circuit board (PCB) or flexible printed circuit (FPC) technologies was utilized to fabricate the electrode boards [12], the performances have been considerably improved, which is now even comparable to electromagnetic motors in some aspects.

Among the macroscale motors, those using FPC technologies realized superior output performance in a linear configuration. For example, the synchronous linear motor reported in [13] provided a large thrust force up to 310 N by stacking 50 pairs of FPC films. For rotary configurations, PCB technology has been favored, whose torque density can be improved by increasing the number of electrodes per unit area/volume [14], [15]. The recent PCB-based prototype in [16] generated a high torque density of 1.4 N·m/kg by using six pairs of double-sided PCBs. In these macroscale motors, synchronous operation principle is commonly used, but various induction principles have also been studied [17], [18], [19].

Macroscale electrostatic motors, especially those using FPC, are expected to be used in robotics, such as climbing robots, humanoid robots, and insect-sized robots, due to their high force-to-weight ratio, lightness, and flexibility. In [20], crawler-type robots with light structure were presented, whose locomotion and adhesion were realized simultaneously by using an electrostatic film motor. In [21], a high-performance muscle-like actuator was reported for humanoid robots, which was fabricated by stacking multiple thin FPC films. The prototype consisted of 11 pairs of films and weighed 235 g, and had mechanical flexibility. Its output force reached 163 N at 1.8 kV, whose force-to-weight ratio reached 694 N/kg. In [22], a new structure for electrostatic film motors with dual-interface actuation was proposed to obtain higher performance in a millimeter-sized actuator. A biologically inspired electrostatic artificial muscle with a maximum output force of 85 m·N and mass of 92 mg was fabricated, which was utilized in a miniature earthworm robot and a compact endoscope surgical tool to achieve tasks in a narrow space.

For applications of macroscale electrostatic motors in robotics and in other applications, a position sensor is always required. There are three position sensing methods that were proposed

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for detecting the slider position of the electrostatic film motors without spoiling their thin structure. In extra-encoder type [12], electrodes for a capacitive linear encoder are printed on the same films as the actuation electrodes. Although the sensing function can be realized on the same films, the extra area required for the sensing electrodes decreases the actuation performance per volume/area. The other two types utilize the actuation electrodes for sensing purpose, while maintaining the actuating function. In high-frequency injection type [23], [24], [25], high-frequency sensing signals are injected onto the low-frequency driving voltages. Then, by detecting the high-frequency current flowing through the actuation electrodes, the slider displacement is obtained in a similar manner as capacitive linear encoders. Although this type can realize high-resolution sensing (when the injection frequency was around 400 kHz), the sensing system is rather complicated for injecting/extracting high-frequency signals onto the high-voltage driving signal. In self-sensing type [26], [27], [28], the slider position information that is coupled in the driving current, could be extracted/distilled by using coordinate transformation methods, such as Clarke and Park transformations. Similar to the self-sensing method of electromagnetic motors [29], [30], [31], the self-sensing method of electrostatic motors has a simple sensing system with low cost. In [27], it was derived under sinusoidal-voltage operation and then was extended to the pulse-voltage operation by eliminating the effects of impulse currents [28]. Theoretical analyzes showed that the proposed self-sensing method could realize high sensing accuracy. However, it cannot work in practical situations due to the considerably small current flowing in the motor. The resulting system had low noise immunity [27] and could not achieve high performance in a practical situation.

To find a suitable application for the self-sensing method, this work focuses on a unique actuation method called *amplitude-modulated ac operation* [32]. In the typical synchronous operation principle, the motor speed is proportional to the applied frequency. Therefore, for low-speed or stationary conditions, the frequency of the applied voltage becomes very low or even becomes direct current (dc). It is reported that the application of dc high voltages causes charging on the insulating film; as the surface of an insulating film has a finite resistance, applying high voltage in the same polarity for a long time will cause a surface current on the film. The resulting charges on the film will disturb the electric field, which causes malfunctioning or loss of thrust force in the electrostatic motor [32], [33]. Such a problem can be avoided by modulating the driving voltage using a high-frequency carrier wave. Due to the modulation, the frequency is always kept high enough even in zero-speed conditions. One drawback of the amplitude-modulated operation is that it requires a larger current due to the use of high frequency. However, the larger current would bring a benefit to the self-sensing. With that in mind, this article discusses position self-sensing in amplitude-modulated ac operation. To facilitate the sensing, this article modifies the self-sensing algorithm in [27], such that it can deal with the amplitude-modulated signals. Although the amplitude-modulated carrier signal voltage has been used for sensorless rotor position estimation in the field of electromagnetic machines decades ago [34], [35],

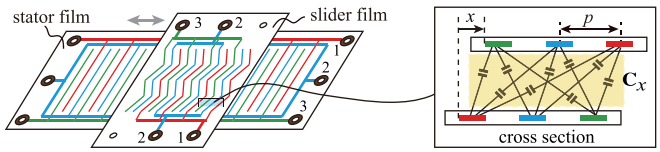


Fig. 1. Structure of an electrostatic film motor with skewed electrodes.

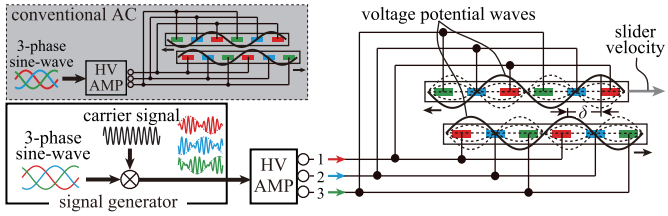


Fig. 2. Principle of amplitude modulated ac operation.

it cannot be directly applicable to electrostatic motors due to their different characteristics from electromagnetic machines. However, it can be used to help modify the self-sensing method under amplitude-modulated ac operation.

The following section introduces the structure of a synchronous-type electrostatic motor briefly and explains the principle of the amplitude-modulated ac operation. In Section III, the driving current under the amplitude-modulated ac voltage is derived analytically, based on which the modified self-sensing algorithm is discussed. Then, simulations for confirming the noise immunity of the modified self-sensing algorithm are carried out using MATLAB/Simulink. The results of verification experiments are introduced in Section IV, and the performance of the modified position self-sensing method is discussed in Section V. Finally, Section VI concludes this article.

II. Synchronous Electrostatic Motor and Amplitude Modulated Operation

A. Structure

Fig. 1 shows an electrostatic linear motor discussed in this work. The motor consists of two films/boards, on which fine-pattern electrodes are fabricated using FPC/PCB technology. In this article, a three-phase variant is focused, in which the stator and the slider have three-phase electrodes, as in the figure. The three-phase electrodes in the stator and the slider are connected, but in the spatially reversed orders, to the same three-phase voltage source, as depicted in Fig. 2.

Capacitance is formed among the electrodes in three phases, which is expressed using a three-by-three symmetric capacitance matrix $\mathbf{C}(x)$ that satisfies

$$\mathbf{Q} = \mathbf{C}(x)\mathbf{V} \tag{1}$$

where $\mathbf{Q}$ and $\mathbf{V}$ are the charge and voltage vectors representing charges and voltages on the three-phase electrodes [34]. Considering the geometrical relations of the electrodes, the capacitance matrix $\mathbf{C}(x)$ can be decomposed into constant and variable

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portions [27], [28] as

$$\mathbf{C}(x) = \mathbf{C}_0 + \mathbf{C}_x(x). \tag{2}$$

The matrix $\mathbf{C}_0$ represents the constant portion of the capacitance matrix and is expressed as

$$\mathbf{C}_0 = \begin{bmatrix} C_{mm} & -C_{mn} & -C_{mn} \\ -C_{mn} & C_{mm} & -C_{mn} \\ -C_{mn} & -C_{mn} & C_{mm} \end{bmatrix} \tag{3}$$

where $C_{mm} = C_{st0} + C_{sl0} - 2C_{m0}$, $C_{mn} = C_{t0} + C_{l0} + 2C_{m0}$, C_{st0}, C_{sl0}, C_{t0}, C_{l0}, C_{m0} are all positive constant depending on the motor structure, and their values were obtained from the actual measured capacitance data by using curve fitting method [27], [28]. The matrix $\mathbf{C}_x(x)$ represents the position-dependent portion and is expressed as

$$\mathbf{C}_x(x) = -2C_{m1}\begin{bmatrix} \cos(\theta_x - \frac{2\pi}{3}) & \cos(\theta_x + \frac{2\pi}{3}) & \cos(\theta_x) \\ \cos(\theta_x + \frac{2\pi}{3}) & \cos(\theta_x) & 2\cos(\theta_x - \frac{2\pi}{3}) \\ \cos(\theta_x) & 2\cos(\theta_x - \frac{2\pi}{3}) & 2\cos(\theta_x + \frac{2\pi}{3}) \end{bmatrix} \tag{4}$$

where C_{m1} is positive constant, $\theta_x = 2\pi x/3p$ is the electrical angle representation of the slider position x, and p is the electrode pitch [27], [28].

B. Amplitude-Modulated AC Operation

In a typical synchronous operation, a set of three-phase ac high voltage is directly applied to the three-phase electrodes in the stator and slider in reverse orders, which generates voltage potential waves running toward opposite directions, as shown in Fig. 2 being enclosed in the gray area. The electrostatic attraction between the two voltage waves creates electrostatic thrust force and propels the slider in a synchronous manner.

In the amplitude-modulated ac operation, on the other hand, the amplitude of the three-phase ac signal is modulated using a high-frequency carrier signal before being applied to the electrodes, as shown in Fig. 2. The three-phase amplitude-modulated sine wave is expressed as

$$\mathbf{V}_{\text{am}} = V_0 \sin(\omega_c t)\left[\sin(\omega t), \sin\left(\omega t - \frac{2}{3}\pi\right), \sin\left(\omega t + \frac{2}{3}\pi\right)\right]^T \tag{5}$$

where V_0 is the amplitude, ω is the angular frequency of the driving voltage, ω_c is the angular frequency of the carrier signal. Normally, the frequencies satisfy $\omega_c \gg \omega$.

The output force can be calculated from the virtual work principle as

$$\begin{aligned} F_{\text{am}} &= \frac{1}{2}\mathbf{V}_{\text{am}}^T \frac{d\mathbf{C}}{dx}\mathbf{V}_{am} = \frac{3\pi}{2p}C_{m1}V_0^2 \sin^2(\omega_c t)\sin\delta \\ &= \frac{3\pi}{4p}C_{m1}V_0^2(1 - \cos(2\omega_c t))\sin\delta \end{aligned} \tag{6}$$

in which $\delta = \theta_x - 2\omega t + \pi/3$ describes the spatial phase difference between the two voltage waves in the stator/slider films (see Fig. 2). Compared with the conventional synchronous operation, the force equation for the amplitude-modulation has an additional coefficient $\sin^2(\omega_c t)$. As a result, the output force vibrates at a high frequency. However, the high-frequency force will be damped in a real system due to the damping effect of a mechanical system. Therefore, the observable output force will be the average force, which can be calculated as

$$\overline{F_{am}} = \frac{\omega_c}{2\pi}\int_0^{2\pi/\omega_c} F_{\text{am}}dt = \frac{3\pi}{4p}C_{m1}V_0^2 \sin\delta. \tag{7}$$

This force equation is the same as the conventional synchronous operation, except for the coefficient, meaning that the motor works as a synchronous motor even under amplitude modulation. If a constant load is given, the output force is kept constant such that it balances with the load, which is realized by keeping δ constant. As $\delta = \theta_x - 2\omega t + \pi/3$ is kept constant, the slider position θ_x will change in accordance with $2\omega t$. In other words, the slider speed is $v = \dot{x} = 6pf$, where $f = \omega/2\pi$ is the frequency of the driving signal.

III. Position Self-Sensing

This work extends the self-sensing algorithm proposed in [27]. Therefore, this section first introduces the algorithm briefly and then explains the modification needed for the amplitude-modulated ac operation.

A. Previous Method

This part briefly introduces the self-sensing algorithm in [27], which was designed for the conventional ac synchronous operation. In the ac operation, the applied voltage is expressed as

$$\mathbf{V}_{\text{AC}} = V_0[\sin(\omega t), \sin(\omega t - 2\pi/3), \sin(\omega t + 2\pi/3)]^T. \tag{8}$$

Then, the driving current, $\mathbf{I}_{\text{AC}}$ is calculated as

$$\begin{aligned} &\mathbf{I}_{\text{AC}} = \frac{d(\mathbf{C}(x)\mathbf{V}_{\text{AC}})}{dt} = A\mathbf{I}_{01} - B\mathbf{I}_{x1} \\ &\begin{cases} A = (C_{mm} + C_{mn})V_0\omega,\ B = 3C_{m1}V_0(\omega_x - \omega) \\ \mathbf{I}_{01} = \begin{bmatrix} \cos(\omega t) \\ \cos(\omega t - \frac{2\pi}{3}) \\ \cos(\omega t + \frac{2\pi}{3}) \end{bmatrix},\ \mathbf{I}_{x1} = \begin{bmatrix} \cos(\theta_x - \omega t + \frac{\pi}{3}) \\ \cos(\theta_x - \omega t - \frac{\pi}{3}) \\ \cos(\theta_x - \omega t + \pi) \end{bmatrix} \end{cases} \end{aligned} \tag{9}$$

where A depends on the angular frequency ω of the applied voltage, and B depends on the slider velocity $\omega_x = d\theta_x/dt$ and the angular frequency ω. By using Clarke and Park transformations, the driving current can be transformed as

$$\mathbf{I}_{\text{AC}dq} = \begin{bmatrix} i_d \\ i_q \end{bmatrix} = A\begin{bmatrix} 1 \\ 0 \end{bmatrix} - B\begin{bmatrix} \cos\delta \\ \sin\delta \end{bmatrix}. \tag{10}$$

From this expression, the spatial phase difference δ can be easily obtained as

$$\delta = \arctan\left(\frac{i_q}{i_d - A}\right). \tag{11}$$

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Using δ, the slider position is calculated as

$$x = \frac{3p}{2\pi}(2\omega t + \delta - \pi/3). \tag{12}$$

B. Driving Current in Amplitude-Modulated AC Operation

In amplitude-modulated operation, the current is also modulated. The modulated current, $\mathbf{I}$, can be obtained by replacing $\mathbf{V}$ in (9) with $\mathbf{V}_{\text{am}}$ as

$$\mathbf{I} = \frac{d(\mathbf{C}(x)\mathbf{V}_{\text{am}})}{dt} = \underbrace{\frac{d(\mathbf{C}_0\mathbf{V}_{\text{am}})}{dt}}_{\mathbf{I}_0} + \underbrace{\frac{d(\mathbf{C}_x(x)\mathbf{V}_{\text{am}})}{dt}}_{\mathbf{I}_x} \tag{13}$$

which can be divided into two components in accordance with the composition of the capacitance matrix $\mathbf{C}(x) = \mathbf{C}_0 + \mathbf{C}_x(x)$. The first component, $\mathbf{I}_0$, has no relation with the slider position and is expressed as

$$\mathbf{I}_0 = \mathbf{C}_0\frac{d\mathbf{V}_{\text{am}}}{dt} = A\sin(\omega_c t)\mathbf{I}_{01} + A_c\cos(\omega_c t)\mathbf{I}_{02}$$
$$\begin{cases} A_c = (C_{mm} + C_{mn})V_0\omega_c \\ \mathbf{I}_{02} = \begin{bmatrix} \sin(\omega t) \\ \sin(\omega t - 2\pi/3) \\ \sin(\omega t + 2\pi/3) \end{bmatrix} \end{cases} \tag{14}$$

in which, A_c depends on the angular frequency ω_c of the carrier signal. Since the carrier frequency is much higher than the driving frequency, i.e., $\omega_c \gg \omega$, the following relation should hold: $A_c \gg A$. The second component in (13), $\mathbf{I}_x$, has relevance to the slider position, which can be calculated as

$$\mathbf{I}_x = \frac{d(\mathbf{C}_x(x)\mathbf{V}_{\text{am}})}{dt} = -B\sin(\omega_c t)\mathbf{I}_{x1} - B_c\cos(\omega_c t)\mathbf{I}_{x2}$$
$$\begin{cases} B_c = 3C_{m1}V_0\omega_c \\ \mathbf{I}_{x2} = \begin{bmatrix} \cos(\theta_x - \omega t - \pi/6) \\ \cos(\theta_x - \omega t - 5\pi/6) \\ \cos(\theta_x - \omega t + \pi/2) \end{bmatrix} \end{cases} \tag{15}$$

where B_c comes from the high-frequency carrier signal with a relation of $B_c \gg B$.

With these variables, the driving current can be rewritten as

$$\mathbf{I} = \sin(\omega_c t)(A\mathbf{I}_{01} - B\mathbf{I}_{x1}) + \cos(\omega_c t)(A_c\mathbf{I}_{02} - B_c\mathbf{I}_{x2}). \tag{16}$$

It can be observed that the high-frequency signals are completely coupled with the low-frequency components in the driving current under amplitude-modulated ac operation. Although the driving current is much more complicated than that in conventional ac operation, its amplitude becomes considerably large, which is an advantage in measuring the current.

C. Modified Self-Sensing Algorithm

In the previous research [27], it was confirmed that Clarke and Park transformations are helpful and effective to simplify the driving current for estimating the slider position. However, they cannot be directly applied to $\mathbf{I}$ in (16).

Considering the high-frequency components of the driving current in the amplitude-modulated ac operation, the driving current needs to be demodulated before being transformed by Clarke and Park transformations. For demodulation, the driving current $\mathbf{I}$ is multiplied by $2\cos(\omega_c t)$ to be $\mathbf{I}_{\cos} = 2\cos(\omega_c t)\mathbf{I}$, which is further expressed as

$$\mathbf{I}_{\cos} = \sin(2\omega_c t)(A\mathbf{I}_{01} - B\mathbf{I}_{x1}) + \cos(2\omega_c t)(A_c\mathbf{I}_{02} - B_c\mathbf{I}_{x2}) + A_c\mathbf{I}_{02} - B_c\mathbf{I}_{x2} \tag{17}$$

in which, the first two terms have high-frequency coefficients, $\sin(2\omega_c t)$ and $\cos(2\omega_c t)$, and the rest has the low frequency of ω. As $\omega_c \gg \omega$, the latter part can be extracted from $\mathbf{I}_{\cos}$ by using a suitable LPF. Ideally, the driving current $\mathbf{I}$ in (16) is demodulated as

$$\mathbf{I}_{\text{dem}} = A_c\mathbf{I}_{02} - B_c\mathbf{I}_{x2}. \tag{18}$$

Mathematically, if $\sin(\omega_c t)$ is used for the demodulation, instead of $\cos(\omega_c t)$, the demodulated driving current in the previous equation becomes $A\mathbf{I}_{01} - B\mathbf{I}_{x1}$. This is identical to the driving current under the conventional ac operation in (9), but this is not of interest due to its small amplitude.

The demodulated driving current $\mathbf{I}_{\text{dem}}$ can be simplified using the Clarke and Park transformations as

$$\begin{cases} \mathbf{I}_{\alpha\beta} = \mathbf{M}_{\alpha\beta}\mathbf{I}_{\text{dem}}, & \mathbf{M}_{\alpha\beta} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{\sqrt{3}}{3} & \frac{2\sqrt{3}}{3} & 0 \end{bmatrix} \\ \mathbf{I}_{dq} = \mathbf{R}_{dq}\mathbf{I}_{\alpha\beta}, & \mathbf{R}_{dq} = \begin{bmatrix} \cos(\omega t) & \sin(\omega t) \\ -\sin(\omega t) & \cos(\omega t) \end{bmatrix}. \end{cases} \tag{19}$$

The simplified expression of the driving current, $\mathbf{I}_{dq}$, can be obtained as

$$\mathbf{I}_{dq} = \begin{bmatrix} i_d \\ i_q \end{bmatrix} = A_c\begin{bmatrix} 0 \\ -1 \end{bmatrix} - B_c\begin{bmatrix} -\sin\delta \\ \cos\delta \end{bmatrix}. \tag{20}$$

From (20), the spatial phase difference δ is easily calculated as

$$\delta = \arctan\left(\frac{-i_d}{i_q + A_c}\right). \tag{21}$$

Since the high-frequency sinusoidal signal $\sin(\omega_c t)$ is introduced into the voltage, components $\mathbf{I}_{02}$ and $\mathbf{I}_{x2}$ were generated in the driving current as in (16). They have a phase difference of $\pi/2$ with $\mathbf{I}_{01}$ and $\mathbf{I}_{x1}$ in (9), respectively. Components $\mathbf{I}_{02}$ and $\mathbf{I}_{x2}$ were remained after demodulation rather than $\mathbf{I}_{01}$ and $\mathbf{I}_{x1}$, and as a result, the position of i_q and i_d in (11) and (21) changed. Then, as same as the previous method [27], the slider position is obtained by using (12). Since the range of δ is limited to $(-\pi/2, \pi/2)$ in arctan calculation, it will be phase-unwrapped by utilizing a phase-unwrapping algorithm, such as in [35] and [36], to obtain the slider position in the range of $(-\infty, \infty)$.

D. Systematic Error Caused by LPF

In the modified self-sensing algorithm, the LPF plays a key role in the demodulation. Ideally, the low-frequency part is preserved perfectly, while the high-frequency parts are completely removed. In reality, however, the high-frequency part cannot be removed completely, which results in a systematic error. The systematic error caused by the LPF is analyzed in the following.

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It is assumed that the gain of the LPF is G_f at the frequency of $2\omega_c$, and it is 1 at the frequency of ω. Due to the relation of $A_c \gg A$ and $B_c \gg B$, the current expression in (17), can be simplified as

$$\mathbf{I}_{\text{in}} = \cos(2\omega_c t)(A_c\mathbf{I}_{02} - B_c\mathbf{I}_{x2}) + A_c\mathbf{I}_{02} - B_c\mathbf{I}_{x2} \tag{22}$$

which is the input of the LPF. The filtered current after the LPF is

$$\mathbf{I}_{\text{out}} = (G_f\cos(2\omega_c t) + 1)(A_c\mathbf{I}_{02} - B_c\mathbf{I}_{x2}). \tag{23}$$

Using the Clarke and Park transformations, the filtered current $\mathbf{I}_{\text{out}}$ will be transformed as

$$\tilde{\mathbf{I}}_{dq} = \begin{bmatrix} i_d \\ i_q \end{bmatrix} = A_c(G_f\cos(2\omega_c t) + 1)\begin{bmatrix} 0 \\ -1 \end{bmatrix} - B_c(G_f\cos(2\omega_c t) + 1)\begin{bmatrix} -\sin\delta \\ \cos\delta \end{bmatrix}. \tag{24}$$

By using (21), tangent of the approximate spatial phase difference $\tilde{\delta}$ is expressed as

$$\tan\tilde{\delta} = \frac{B_c(G_f\cos(2\omega_c t) + 1)\sin\delta}{A_c G_f\cos(2\omega_c t) + B_c(G_f\cos(2\omega_c t) + 1)\cos\delta}. \tag{25}$$

The error between $\tilde{\delta}$ and δ is expressed as $\Delta\delta = \tilde{\delta} - \delta$, whose tangent is calculated as

$$\begin{cases} \tan\Delta\delta = \dfrac{-A_c G_f\cos(2\omega_c t)\sin\delta}{A_c G_f\cos(2\omega_c t)\cos\delta + B_c(G_f\cos(2\omega_c t) + 1)} \\ \lim_{G_f\to 0}\tan\Delta\delta = 0. \end{cases} \tag{26}$$

When G_f is small enough, the denominator becomes B_c, and the error $\Delta\delta$ is approximated as

$$\Delta\delta \approx \arctan\left(\frac{-A_c G_f}{B_c}\sin\delta\cos(2\omega_c t)\right). \tag{27}$$

This equation confirms that the error $\Delta\delta$ depends on the remained high-frequency components at $2\omega_c$, and its magnitude is in a range of $[\arctan(-A_c G_f/B_c), \arctan(A_c G_f/B_c)]$. The slider position that is calculated by substituting $\tilde{\delta}$ into (12) is denoted as $\tilde{x}$. Then, the systematic error, $\varepsilon_s = \tilde{x} - x$, of the calculated position is expressed as

$$\varepsilon_s = \frac{3p}{2\pi}\Delta\delta,\ |\varepsilon_s| \leqslant \frac{3p}{2\pi}\arctan\left(\frac{A_c G_f}{B_c}\right). \tag{28}$$

The maximum estimation error depends on the performance of the LPF at the frequency of $2\omega_c$ and the capacitance coefficients of the motor. Smaller cut-off frequency results in smaller G_f, as well as smaller the systematic error ε_s, but it will also limit the bandwidth of the sensor output.

E. Noise Immunity

The self-sensing method in this work is expected to have higher noise immunity than the previous method in [27]. In this work, high noise immunity means that the signal-to-noise ratio (SNR) is higher in the same noisy environment, or the self-sensing method can work well under lower SNR conditions. On the contrary, it is low noise immunity. In the conventional ac operation, the driving current is very small and the coefficient B is susceptible to the slider velocity ω_x, which resulted in low noise immunity. On the other hand, the driving current in the amplitude-modulated ac operation is amplified by the high-frequency carrier signal, which could enhance the noise immunity of the self-sensing algorithm.

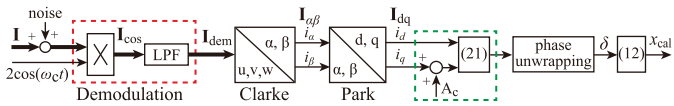


Fig. 3. Block diagram of the modified self-sensing algorithm, which was used to evaluate its noise immunity in simulations.

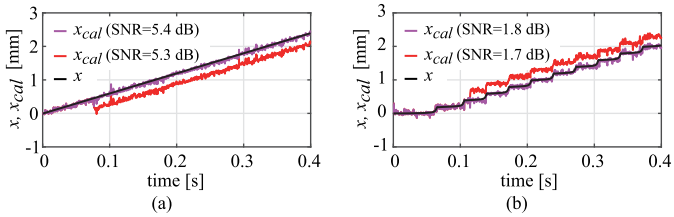


Fig. 4. Simulation results in different motors when noise was added to the driving current ($f = 5$ Hz, $f_c = 1$ kHz, $f_{\text{cutoff}} = 0.5$ kHz).

To evaluate the performance of the modified self-sensing method in a noisy environment, numerical simulations were carried out using MATLAB/Simulink R2020b. The block diagram of the electrostatic motor and all the parameters were adopted from [27], except for those related to the amplitude modulation. The amplitude of $\mathbf{V}_{\text{am}}$ in (5) was set to 1.5 kV, and the driving frequency of $f = \omega/2\pi = 5$ Hz was used, for simulating the dynamic motion. The carrier wave frequency $f_c = \omega_c/2\pi$ was changed between 0.1 kHz and 1 kHz and the slider behavior as well as the driving current was calculated in Simulink. The capacitances used for the simulations are

$$\begin{cases} C_{st0} = 540, C_{sl0} = 314, C_{t0} = 180, C_{l0} = 67 \\ C_{m0} = 60, C_{m1} = 50, C_{mm} = 734, C_{mn} = 367 \end{cases} \tag{29}$$

in units of [pF].

The simulated driving current was fed into the block diagram of the modified self-sensing algorithm shown in Fig. 3, which was also implemented in Simulink. The area enclosed by dashed lines shows the differences from the previous method in [27]. The red dashed box represents the demodulation block, while the green dashed box represents the calculation of spatial phase difference δ using (21). A band-limited white noise block was used to add noise into the driving current $\mathbf{I}$, before being fed to the demodulation block. A third-order Butterworth LPF filter was used in demodulation, whose cutoff frequency was $f_{\text{cutoff}} = 0.5 f_c$. The sampling period of the simulation was set to 10 μs (100 kHz).

Fig. 4 shows the simulation results with noise for two different motors, in which the carrier frequency was $f_c = 1$ kHz. The two motors differ in the shape of the slider electrodes; one is

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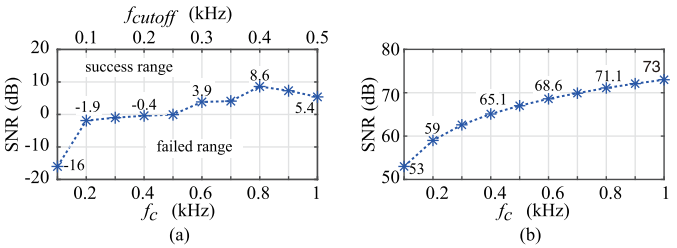


Fig. 5. SNRs under various conditions (skewed motor, $f = 5$ Hz). (a) Smallest SNRs for successful estimation at different carrier frequencies ($f_{\text{cutoff}} = 0.5 f_c$), (b) SNRs at different carrier frequencies, when the same noise of SNR $= 30$ dB in [27] was added to the driving current.

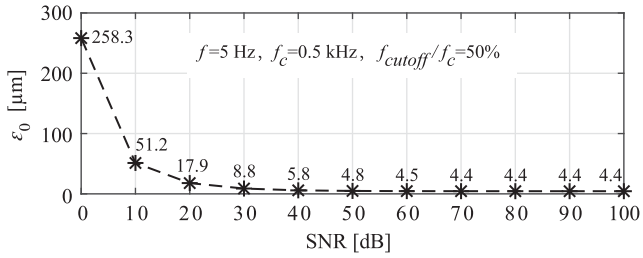


Fig. 6. Relation between accuracy and SNRs of the proposed self-sensing method in a skewed motor ($f = 5$ Hz, $f_c = 0.5$ kHz, $f_{\text{cutoff}} = 0.5 f_c$).

skewed and the other is nonskewed, which is referred to as a regular motor. While the skewed motor has an ideal capacitance matrix, the regular nonskewed motor has distortion in the capacitance that causes stepwise motions [34]. As the self-sensing method was proposed by using the skewed motor, there will be a theoretical error for the regular motor due to the capacitance constant C_{m2} [28]. When the slider position of a regular motor is estimated by using this presented method, the error will be larger than that for a skewed motor, which must be accounted for. However, extensive simulation data implied that the self-sensing capability of this method does not vary upon the load condition.

In the figure, black lines represent the true slider positions x, which were numerically simulated considering the dynamics of the motor. Purple and red lines show the estimated slider positions x_{cal} under different noise levels using the modified self-sensing algorithm. In the case of the skewed motor, the modified self-sensing method worked well when SNR ≥ 5.4 dB, as shown in Fig. 4(a) using the purple line. However, when SNR < 5.4 dB, a fatal error, i.e., jump in the output, was observed as shown in (a) using the red line. For the regular motor, the modified self-sensing method can track the slider position well when SNR ≥ 1.8 dB, while it failed when SNR < 1.8 dB, as shown in Fig. 4(b).

In the previous method for the conventional ac operation in [27], the self-sensing can only work well when SNR was more than 59 dB, which was almost impossible to be satisfied in real situations. Also in [27], the threshold for stable operation was improved to 18 dB, by compromising the resolution. Compared with those previous results, the modified self-sensing has far higher noise immunity. The high noise immunity of the modified method mainly comes from the demodulation. As the LPF in demodulation cuts down the noise far from the driving frequency, it considerably reduces the noise power and improves SNR during signal processing. Fig. 5(a) shows the lowest SNR to realize successful sensing at various carrier frequencies. The cut-off frequency of the LPF was set to half of the carrier frequency. As the cut-off frequency is proportional to the carrier frequency, higher noise immunity was obtained at a lower carrier frequency due to the stronger LPF. This indicates that LPF in the demodulation process is also contributing to noise immunity.

In addition, the driving current is much larger in amplitude-modulated operation, which makes the SNR much higher against the same noise level. For example, Fig. 5(b) shows SNRs when a noise with a constant level is applied. In [27], the same noise level resulted in SNR $= 30$ dB for an ac driving signal. In the modified method, on the other hand, SNR was 53 dB, even when the lower carrier frequency of 100 Hz is utilized. Because of these two reasons, the modified self-sensing method has much higher noise immunity than the previous method in [27], which makes this modified method practical enough, as is demonstrated in the following section.

To clarify how noise affects the accuracy of this self-sensing method, simulations were also carried out in environments with different noise levels, in which the carrier frequency was $f_c = 0.5$ kHz and SNR changed from 0 to 100 dB with an interval of 10 dB. The estimation error ε is defined as the difference between the estimated slider position x_{cal} and the true slider position x, which is $\varepsilon = x_{\text{cal}} - x$. Its amplitude, which is expressed as $\varepsilon_0 = |\varepsilon_{\max} - \varepsilon_{\min}|/2$, was used to evaluate the estimation error of the self-sensing method. The relation between the estimation error ε_0 and the noise level is shown in Fig. 6. It confirmed that the higher the noise level is, the greater the estimation error becomes. When the noise was reduced to a very low level (SNR < 60 dB), there was an error constant, which is the systematic error caused by the LPF during demodulation. This systematic error could be reduced by adjusting the cutoff frequency f_{cutoff}, which could be confirmed in the following section.

IV. Experimental Verification

In order to evaluate the performance of the modified self-sensing method in a practical situation, experiments were performed under various conditions, including different electrode configurations, different cut-off frequencies of the LPF in demodulation, and different carrier frequencies.

A. Experiment Setup

Fig. 7 shows the experiment setup, in which two different motors were utilized. In both motors, the electrodes in the stator and the slider were aligned with a pitch of 100 μm. The stator films shared the same design in the two different motors. The slider films were different in the two motors, especially in electrode shape. In the skewed motor, the slider electrodes had

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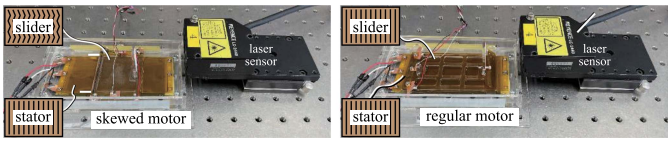


Fig. 7. Experiment setup.

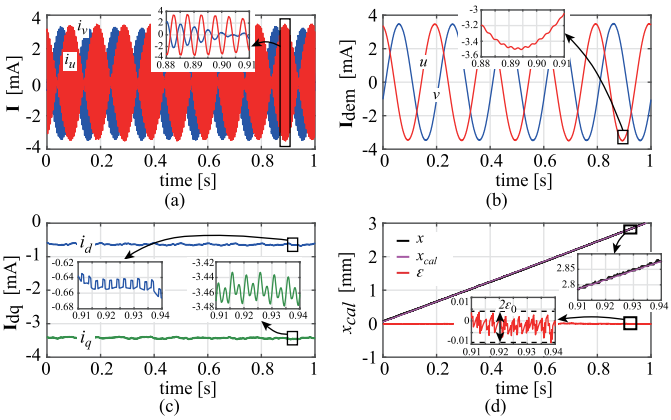


Fig. 8. Signal processing of the modified self-sensing method in case of a skewed motor ($f = 5\,\text{Hz}$, $f_c = 200\,\text{Hz}$, $f_{\text{cutoff}} = 0.3f_c$). (a) driving current $\mathbf{I}$. (b) Demodulated current $\mathbf{I}_{\text{dem}}$. (c) Simplified current $\mathbf{I}_{dq}$. (d) Calculated position x_{cal}.

a zig-zag shape, as shown in the figure. On the other hand, the regular motor had straight electrodes in the slider. This slider film had the same design as the stator film, but was shortened in length by bending both ends. In both motors, the stator film was fixed to a plastic box, and the slider displacement was measured. The motors were operated in a dielectric fluid (3M, Flourinert FC-3283) to prevent discharge in air, and glass beads with a diameter of 20 μm were sprinkled between films to reduce friction.

A three-phase amplitude-modulated ac voltage was generated by using a digital signal processing system (Dspace, ds1104, sampling frequency: 10 kHz) and then was amplified by a set of high-voltage amplifiers (NF, HVA4321) before being applied to the motors, as shown in Fig. 2. The amplitude of the applied voltage was constant ($V_{0-p} = 1\,\text{kV}$), while the frequencies f and f_c changed among different experiments. A laser displacement sensor (KEYENCE, LC-2440) was used to measure the slider position as reference. The driving current was measured using the built-in current monitor in the high-voltage amplifiers, which has an accuracy of 50 μA. The output of the current monitor was recorded at a sampling frequency of 10 kHz.

B. Results

For evaluating the performance of the self-sensing method under various conditions, the current was processed offline, using the recorded current signals. Fig. 8(a) shows an example of the recorded driving current, $\mathbf{I}$, which is for the skewed motor driven with $f = 5\,\text{Hz}$ and $f_c = 200\,\text{Hz}$. For demodulation, a third-order Butterworth LPF with a cutoff frequency of

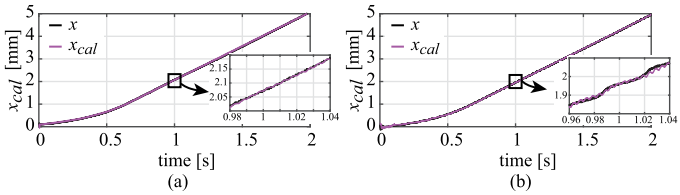


Fig. 9. Estimated slider position in different motors when the driving frequency changed from 0 to 5 Hz ($f_c = 500\,\text{Hz}$, $f_{\text{cutoff}} = 0.3f_c$). (a) Skewed motor. (b) Regular motor.

$f_{\text{cutoff}} = 0.3f_c$ was used. The demodulated current $\mathbf{I}_{\text{dem}}$ is shown in Fig. 8(b), in which the remained high-frequency signals can also be observed. After Clarke and Park transformations, $\mathbf{I}_{\text{dem}}$ was simplified to $\mathbf{I}_{dq}$, as shown in Fig. 8(c). Then, the spatial phase difference δ was calculated by using (21), in which A_c was set as 2.64 mA. By using the phase-unwrapped δ and the phase of the applied voltage, ωt, the slider position x_{cal} was calculated using (12), as shown in Fig. 8(d) using the purple line. For comparison, the real slider position x, which was measured by the laser sensor, was also plotted in the same figure using the black line. The red line indicates the estimation error $\varepsilon = x_{\text{cal}} - x$. It is observed that ε has a high frequency, which was caused by the remained high-frequency component in $\mathbf{I}_{\text{dem}}$. Its amplitude, which is expressed as $\varepsilon_0 = |\varepsilon_{\max} - \varepsilon_{\min}|/2$, was used to evaluate the estimation error of the self-sensing method.

To compare the performance of the modified self-sensing method in different motors, experiments were performed under various conditions. A three-phase voltage with a constant carrier frequency f_c and a variable driving frequency f was applied to the skewed motor and the regular motor, respectively. The motor was accelerated by changing the driving frequency f from zero to 5 Hz, and the speed was kept constant from 0.6 s, by fixing f, as in Fig. 9.

The motors were operated under different carrier frequencies with an interval of 100 Hz. For each carrier frequency, the slider position was estimated offline under different cut-off frequencies with an interval of $f_{\text{cutoff}} = 0.1f_c$ to evaluate how the LPF affects the estimation error. The purple line in Fig. 9 shows an example of the estimated slider position, in which the carrier frequency was $f_c = 500\,\text{Hz}$ and the cutoff frequency was $f_{\text{cutoff}} = 0.3f_c$. The black line indicates the real slider positions measured by the laser sensor. The plots show that the skewed motor moved smoothly at a constant speed, while the regular motor moved stepwise, which is consistent with the simulation. In both motors, in this case, the modified self-sensing method successfully tracked the slider position without fatal errors.

Fig. 10 shows the relation between the cutoff frequency f_{cutoff}, the carrier frequency f_c, and the estimation error ε_0. When $f_c = 500\,\text{Hz}$, the error ε_0 monotonically increased as the cutoff frequency was increased, for both the skewed motor and the regular motor, as shown in Fig. 10(a). Fig. 10(b) shows the errors for different carrier frequencies. The noise was found to decrease up to the carrier frequency of 400 Hz. However, the error ε_0 increased when $f_c = 500$ Hz. This was because the cut-off frequency of LPF was changed in proportion to the

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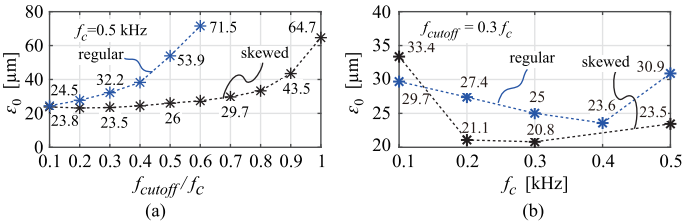


Fig. 10. Estimation error ε_0 under various conditions (different carrier frequencies f_c, different cutoff frequencies f_{cutoff}, different motors. (a) error ε_0 under different f_{cutoff}/f_c. (b) Error ε_0 under different f_c.

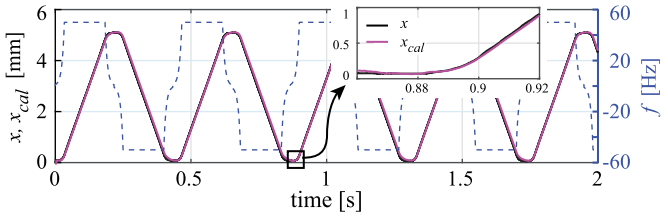


Fig. 11. Estimated slider position in reciprocating movements of the skewed motor ($f_c = 0.5$ kHz, $f_{\text{cutoff}} = 0.3 f_c$). The motor was driven in an open loop and the sensor signal was processed online.

carrier frequency. The higher cut-off frequency allowed much more noise to survive the filtering, which affected the final result. When comparing the two motors, the error in the regular motor was found larger. This is because the driving currents were distorted by the capacitance distortion and the resulting stepwise movement.

In Fig. 11, the motor was reciprocated in the range of [0, 5] mm. The driving frequency in the region with a constant speed was either -50 Hz or 50 Hz. When turning, the frequency was gradually changed between the two frequencies, as indicated by the dashed blue line. The carrier frequency was $f_c = 500$ Hz, and the cut-off frequency was $f_{\text{cutoff}} = 0.3 f_c$. The plot confirms the repeatability of the self-sensing method in a reciprocating motion.

V. Discussion

Compared with the previous method in [27], the modified self-sensing method in this work has many advantages. To clarify the performance of the previous method in [27] for comparison with this work, the results are compared for conventional ac operation and amplitude-modulated ac operation, respectively, as shown in Fig. 12. Fig. 12(a-1) shows the driving current under conventional ac operation, in which the driving frequency f changed from zero to 5 Hz and it was fixed from 0.6 s, while the real slider position x is plotted in Fig. 12(a-2) using black line. For comparison, the driving current under the amplitude-modulated ac operation that corresponds to Fig. 9(a) was plotted in Fig. 12(b-1), and the real slider position x was plotted in Fig. 12(b-2) using the black line. From Fig. 12(a-1), it was observed that when the motor speed was slow, the driving current became considerably small under conventional ac operation,

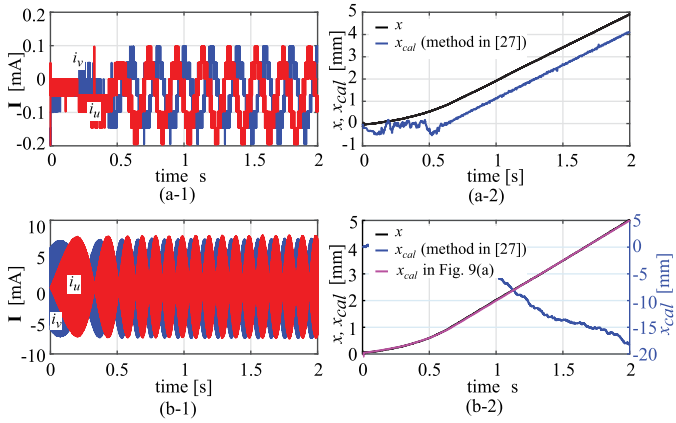


Fig. 12. Position of the skewed motor was estimated by using the previous method in [27], when the motor was driven under conventional ac operation and amplitude-modulated ac operation, respectively.

which was almost impossible to measure. On the other hand, the driving current in this work is easily measured due to its large amplitude, as shown in Fig. 12(b-1).

Since the previous method has considerably low noise immunity, it could not be directly used in a practical environment. To cope with the noise problem, the previous work also proposed the compromised method (which was called "method 2" in [27]), which could be used in a practical environment. By feeding the driving current as in Fig. 12(a-1) and (b-1) to the compromised method, the slider position x_{cal} can be estimated, which was plotted by using the blue lines in Fig. 12(a-2) and (b-2), respectively. For comparison, the estimated slider position using the modified self-sensing method shown in Fig. 9(a) was also plotted in Fig. 12(b-2) using purple line. In the signal processing, a high-pass filter (third-order Butterworth, cutoff frequency 10 Hz) was used to remove the base wave, and an LPF (third-order Butterworth, cutoff frequency 150 Hz) was used to remove noise. From Fig. 12(a-1) and (a-2), it was observed that the previous method could estimate the slider position only when the speed is at a high level, while it cannot track the slider position at a low speed due to the undetectable current. However, it totally failed to track the slider position under the amplitude-modulated ac operation, as shown in Fig. 12(b-2), because there is no suitable strategy to deal with the high-frequency component in the driving current. On the other hand, the modified self-sensing method in this work can track the slider position even at a low speed, and it could extract the slider information from the driving current in the amplitude-modulated ac operation.

Because of the large driving current, the accuracy is much higher than the previous method in [27]. In [27], the compromised method was used in a practical environment to replace the previous method due to its low noise immunity. However, its resolution was compromised to one electrode pitch. On the other hand, the modified self-sensing method in this work achieved much higher accuracy. Although not explicitly shown, our simulation results indicated that the error can be smaller than 1 μm, for a motor with 100-μm electrode pitch. Although

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the accuracy was degraded in the actual experiments due to the performance of the digital signal processing system and the influence of the noise in the driving current, it still reached a low level of 20 μm. Also, it is expected to be improved by increasing the carrier frequency.

In future applications, the carrier frequency f_c and the cut-off frequency f_{cutoff} should be determined carefully. Experimental results confirmed that higher carrier frequency leads to better resolution. However, the higher carrier frequency may lead to a larger dielectric loss in the motor. Also, it may require another special amplifier that has a wider (or higher) bandwidth. The cut-off frequency f_{cutoff} in demodulation can also realize better resolution. However, a lower cut-off frequency will limit the output bandwidth of the self-sensing. Therefore, the carrier frequency and the cutoff frequency of the LPF should be determined according to the requirements of the real system.

Considering the limitation of power supply and high dielectric loss, there is no need to use high-frequency current at all speeds in future applications. To balance the high dielectric loss and noise sensitivity in reality, a feasible strategy is to use this modified method at low speed and switch it to the previous method [27] at high speed. However, the switching condition is not easy to be determined, which should be adjusted according to the performance of each motor and each power supply in reality with the consideration of the accuracy requirement.

VI. Conclusion

This work improved a position self-sensing method for electrostatic motors under the amplitude-modulated ac operation, which can achieve high accuracy with high noise immunity. Although the driving current has a very complicated expression, the position information can be distilled by combining a demodulation algorithm with the Clarke and Park transformations. This article discussed a systematic error ε_s caused in the filtering process during demodulation. The theoretical analysis implied that the systematic error is reduced by lowering the cut-off frequency. Simulation results indicated that noise immunity can be increased in amplitude-demodulated ac operation due to the high-frequency carrier signal and the LPF, which was much better than the previous method. Experimental results confirmed the accuracy, practicality, and stability of this sensing method in several different conditions. It was verified that the slider position could be measured with an accuracy of around 20 μm, which is better than the previous method. Although the accuracy was limited by the performance of the digital signal processing system and the power supply used in the experiments, it is expected to be improved by using different components in the future.

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